

Roll No.....

ODD MID Term Examination-September (2025)

Btech Electrical 3rd Semester

Electrical and electronic measuring instruments (TEE 303)

Time: 1:30 Hrs

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO-01
(a)	Explain direct and indirect measurements. Also explain null type instruments.	
(b)	A moving coil instrument gives a full scale deflection of 10 mA when the potential difference across its terminals is 100 mV. Calculate (i) the shunt resistance for a full scale deflection corresponding to 100A, (ii) the series resistance for full scale reading with 1000V. Calculate the power dissipation in each case.	
Q.2	(10 marks)	CO-02
(a)	In a Schering bridge, the arms have the following values: - Arm1: Capacitor $C_1 = 500 \text{ pF}$ in parallel with resistor $R_1 = 1000 \Omega$ - Arm2: Standard capacitor $C_2 = 1000 \text{ pF}$ - Arm 3: Resistor $R_3 = 200 \Omega$ Find the values of unknown capacitance (C_x) and resistance (R_x).	
(b)	Explain the construction and working principle of an electrodynamic type wattmeter. Derive the torque equation for it.	
Q3	(10 marks)	CO-05
(a)	Explain principle and working of PMMC instruments with relevant diagram.	
(b)	Explain two wattmeter method of power measurement in three phase wye connected system.	
Q4	(10 marks)	CO1
(a)	Explain how De Sauty's bridge helps in comparing two capacitances.	

(b)	Draw and explain the Maxwell's Bridge for the measurement of inductance. In a Maxwell's Bridge, the following values are used: $R_1 = 100 \Omega$, $R_2 = 500 \Omega$, $R_3 = 400 \Omega$, $C = 1 \mu F$ Find the unknown inductance (L) and resistance (R) of the coil.	
Q5	(10 marks)	
(a)	Explain series type ohmmeter. Derive expressions of resistances involved.	CO5
(b)	A balanced star connected load of $(8+j6)\Omega$ per phase is connected to a balanced 3-phase 400 V supply. Find the line current, power factor, power and total volt amperes.	